

Development of Multifunctional Carboxymethyl Starch-Based Semi-IPN Hydrogels for Water and Fertilizer Management in Maize Cultivation under Different Water Deficit Conditions

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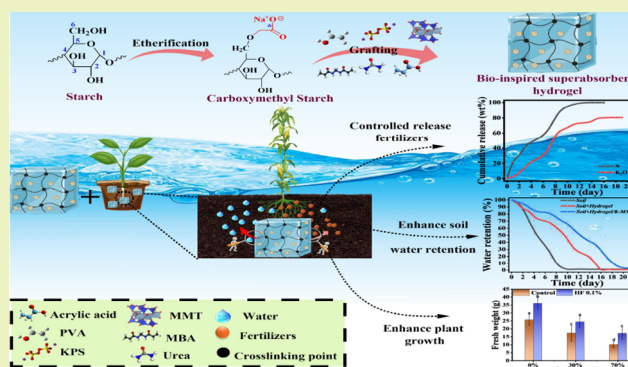
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ABSTRACT: Agricultural sustainability faces significant challenges from water scarcity, inefficient fertilizer use, and environmental pollution, all of which threaten food security and ecosystem health. Conventional fertilizers often fail to synchronize nutrient release with crop demand, leading to substantial losses through leaching and runoff, while drought stress exacerbates these inefficiencies by limiting nutrient uptake. Addressing these critical issues requires innovative materials capable of enhancing water retention, minimizing nutrient loss, and supporting crop productivity under adverse conditions. This study introduces a novel, cost-effective, and sustainable hydrogel-based nanocomposite fertilizer carboxymethyl starch-g-polyacrylic acid/PVA/K-MMT@Urea designed to tackle these agricultural challenges. Synthesized via a simple radical in situ copolymerization process, the composite exhibits exceptional water absorption capacities (392.87 ± 8.05 g/g in distilled water and 70.38 ± 4.26 g/g in 0.1 M NaCl solution) and integrates slow-release mechanisms for nitrogen and potassium. Incorporating 1% of the hydrogel into soil significantly improved its water-holding capacity, extending moisture availability during drought conditions. When applied to maize cultivation, the hydrogel improved key growth parameters including leaf number, chlorophyll content, stem diameter, fresh and dry biomass, and nutrient uptake under drought stress. This dual-action material, combining water conservation and slow nutrient release, demonstrates its potential as a transformative solution for improving crop yield and reducing environmental impacts in water-limited agriculture.

KEYWORDS: starch, bioinspired superabsorbent, hydrogels, controlled-release fertilizers, soil water retention, sustainable agriculture



1. INTRODUCTION

By 2050, global food demand is expected to increase dramatically, requiring a doubling of food production.^{1,2} However, meeting this demand sustainably and environmentally friendly represents a major hurdle for the agricultural industry, as it uses approximately 70% of the world's available water supply for human use.³ Furthermore, the challenges facing the agricultural sector in the coming years include reduced water availability, droughts, and climate change, which could result in a decrease in agricultural production of 2–15% over the next 30 years.^{4,5} Therefore, it is crucial to find innovative solutions to manage water resources efficiently and ensure the long-term sustainability of the agricultural industry. In recent years, the use of superabsorbent hydrogels (SAHs) in agricultural and horticultural applications has gained increasing interest due to their potential to improve crop survival and reduce drought stress in arid and semiarid regions.⁶ SAHs are three-dimensional cross-linked polymer networks that can absorb and retain large amounts of water and salt solutions

from tens to thousands of times their own weight while holding them under pressure.

Another critical challenge in agriculture is fertilizer loss, with up to 70% of applied nitrogen and 80–90% of phosphorus fertilizers lost to the environment, leading to greenhouse gas emissions, water contamination, and soil degradation.^{7,8} Addressing this issue requires strategies that optimize nutrient availability while minimizing environmental impact. Controlled-release fertilizers (CRFs) are a promising solution, as they provide nutrients in a controlled manner, aligning with plant uptake needs and reducing losses.⁹ By incorporating SAHs into CRFs, nutrient efficiency can be further enhanced,

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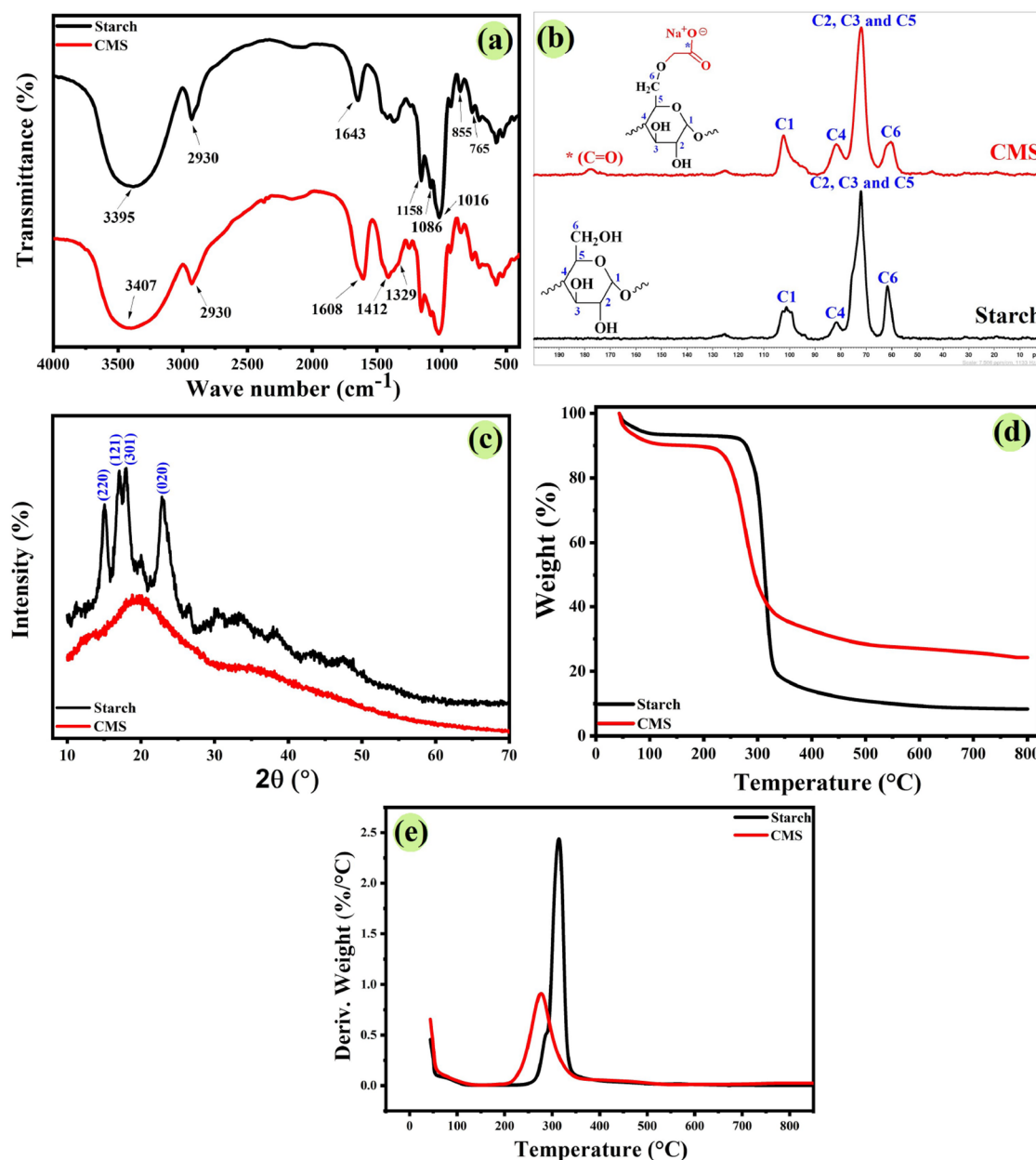


Figure 1. Comprehensive characterization of starch and CMS: (a) FTIR, (b) solid-state ^{13}C NMR, (c) XRD, (d) TGA, and (e) DTG.

ensuring a steady supply of essential elements while reducing environmental impact.¹⁰ Beyond their role in nutrient management, SAHs improve soil structure by enhancing porosity, permeability, and water infiltration while reducing runoff and erosion. Their ability to absorb and retain moisture minimizes irrigation losses, providing a sustained water supply even under drought conditions. Additionally, SAHs promote oxygen circulation in the soil, further supporting root development and plant growth.^{11–13} Natural polymers have emerged as a preferred solution in agricultural settings, owing to their inherent characteristics such as biocompatibility, biodegradability, and cost-effectiveness. Natural polymers have attracted considerable interest across various fields due to their biocompatibility, biodegradability, and cost-effectiveness. They are widely applied in drug delivery, tissue engineering, environmental remediation, and agriculture.^{14–16} Among these, starch; a renewable biopolymer composed of linear amylose and branched amylopectin stands out for its

versatility and abundance. Its physicochemical properties can be tailored based on the amylose-to-amylopectin ratio, which varies depending on botanical origin. Beyond its nutritional value, starch shows great promise in nonfood sectors, particularly in the development of SAHs.¹⁷ To enhance starch's hydrophilicity and functionality, carboxymethylation is widely employed, yielding carboxymethyl starch (CMS). This derivative is commonly used in food, pharmaceutical, and agricultural systems due to its improved water solubility.^{18–20} However, CMS alone exhibits limited water absorbency and mechanical strength, which constrains its independent use in hydrogel applications. To overcome these limitations, researchers have explored grafting hydrophilic vinyl monomers onto CMS and blending it with synthetic polymers. One of the most effective strategies is the development of semi-interpenetrating polymer network (semi-IPN) hydrogels. Semi-IPN hydrogels form through the entanglement of a linear polymer within a cross-linked polymer network, without any covalent

interactions between them. This structure enhances both mechanical strength and thermal stability of the hydrogel.²¹ In this context, poly(vinyl alcohol) (PVA) is commonly used due to its water solubility, hydrophilicity, biodegradability, and excellent compatibility with natural polymers. Incorporating PVA into semi-IPN systems significantly improves swelling behavior, chemical resistance, and the overall durability of the hydrogel matrix.²² Further performance enhancement can be achieved by incorporating nanofillers such as metal oxide nanoparticles, biochar, graphene oxide, and natural clays. Among these, montmorillonite (MMT); a layered aluminum silicate with high aspect ratio, ion-exchange capacity, and reactive surface has demonstrated outstanding potential. MMT significantly improves the mechanical integrity, thermal stability, swelling behavior, and nutrient adsorption efficiency of hydrogels.²³

Numerous studies have explored the potential agronomic advantages of utilizing biopolymers based SAHs or SAHs nanocomposites under conditions of reduced irrigation, as well as the favorable effects of such materials on various indicators of plant growth.^{24–26} For instance, Sharma et al.²⁷ developed a biodegradable, superabsorbent N–S–P carbon dots-enriched carboxymethylcellulose (CDs@CMC) nanohydrogel to enhance crop growth under limited water conditions. Characterization confirmed its structure, and soil studies demonstrated improved swelling (up to 2712%), water-holding (41.2%), and retention capacities, extending soil moisture from 5 to 12 days. Pot experiments with Vigna radiata used 1 wt % of Gel-1, Gel-2, and Gel-3 mixed with 50 g of soil, showing improved plant survival. Gel-2 performed best due to its balanced N–S–P CDs composition. Furthermore, Liu et al.²⁵ developed chitosan-grafting-poly(acrylic acid)/green tea/calcined oil shale semicoke (CS-g-PAA/GT/COSSC) superabsorbent composites with superior water absorption and retention properties. The composites exhibit high water absorption in distilled water and NaCl solution and show good biodegradability due to the presence of natural polymer residues. The study also shows that the composites have excellent soil water holding capacity and can promote cabbage seed growth under water-limited conditions, as evidenced by increased seed biomass and stem length.

Despite the excellent properties of superabsorbent in agriculture, most studies primarily focus on water retention while neglecting their role in fertilizer delivery. Additionally, reported works often use more than 0.5% hydrogels in pot experiments, which may negatively impact soil health over time. Our objective is to develop a superabsorbent fertilizer that enhances both water retention and controlled nutrient release while assessing its performance at a lower, sustainable concentration of 0.1%. To the best of the authors' knowledge no prior research has examined the application of a semi-IPN SAH based on CMS, PVA, and acrylic acid, reinforced with potassium-enriched montmorillonite (K-MMT), as a soil amendment for greenhouse maize cultivation under various water stress conditions. In this study, K-MMT acts as a multifunctional nanofiller and cross-linker, enhancing the hydrogel's mechanical properties, cross-linking density, and water absorption capacity, even in small quantities, by forming a highly cross-linked polymer network. This research is part of our broader program aimed at developing innovative, sustainable, and cost-effective solutions to enhance water and nutrient use efficiency in crops.^{28–34} The hydrogel is synthesized via graft copolymerization of AA onto a CMS

backbone, facilitated by PVA in an aqueous medium, incorporating MBA as a cross-linker, KPS as an initiator, K-MMT as a nanofiller, and urea as a nitrogen source. Urea integration aims to enhance the hydrogel's slow-release properties for nitrogen and potassium. Beyond synthesis and characterization, this study assesses the hydrogel's impact on maize growth under different water stress conditions, demonstrating its potential to enhance drought resilience, nutrient uptake, and sustainable agriculture.

2. EXPERIMENTAL SECTION

The preparation details of CMS, the semi-IPN hydrogel, and its nanocomposite, along with swelling measurements, deswelling kinetics in saline solutions, reswelling experiments, water retention capacity, nitrogen and potassium release behavior, pot culture experiments on maize plants, and swelling kinetics are provided in the Supporting Information.

3. RESULTS AND DISCUSSION

3.1. Characterization of Carboxymethyl Starch. The FTIR spectra of native and carboxymethylated starch are presented in Figure 1a. In native starch, the broad O–H stretching band appears at 3395 cm^{−1}, while the peak at 2930 cm^{−1} corresponds to C–H stretching. The absorption at 1643 cm^{−1} is attributed to O–H scissoring of water molecules, and the characteristic C–O stretching bands at 1158, 1086, and 1016 cm^{−1} distinguish organized and amorphous starch structures.^{35,36} Skeletal vibrations are observed at 855 and 765 cm^{−1}. In CMS, the spectral profile between 700 and 1200 cm^{−1} is retained, while the emergence of bands at 1608, 1412, and 1329 cm^{−1} confirms the introduction of carboxylate (–COO[−]) groups. The intensified broad O–H stretching (3600–3000 cm^{−1}) and CH₂ symmetric stretching (2930 cm^{−1}) further confirm successful carboxymethylation.³⁷

To further support the findings from FTIR, the etherification of maize starch was analyzed via NMR spectroscopy. Figure 1b presents the ¹³C CP/MAS NMR spectra of native and carboxymethylated starch. The spectra of native starch exhibits four main signals corresponding to the anhydroglucose unit, with the C-4 glycosidic bond carbons at 81.5 ppm, indicative of amorphous regions.³⁸ Signals between 68–78 ppm correspond to C-2, C-3, and C-5, while the peak at 61.4 ppm is assigned to C-6. The C-1 carbons appear in the 99–103 ppm range, where the triplet pattern at 99.5, 100.5, and 101.5 ppm confirms the A-type crystallinity of maize starch.³⁹ Following carboxymethylation, a new signal at 178.01 ppm confirms the introduction of carbonyl (C=O) groups. Additionally, reduced intensity of the C-6 and C-4 signals suggests partial degradation of crystalline regions, further validating the structural modification.

Figure 1c presents the XRD spectra of native and carboxymethylated corn starch. Native starch exhibits a semicrystalline structure with characteristic A-type peaks at 2θ values of 15.07°, 17.08°, 17.97°, and 22.98°. ⁴⁰ Carboxymethylation significantly reduces crystallinity due to (i) heat-induced granule disruption in an aqueous medium, (ii) conversion of hydroxyl (St–OH) groups into alkoxide (St–O[−]) under alkaline conditions, generating repulsive forces that disrupt the crystalline network, and (iii) hydrogen bond disruption caused by hydroxyl substitution with carboxymethyl groups.^{41–43} This reduction in crystalline enhances the amorphous nature of CMS, improving its water absorption capacity, which is advantageous for hydrogel applications.

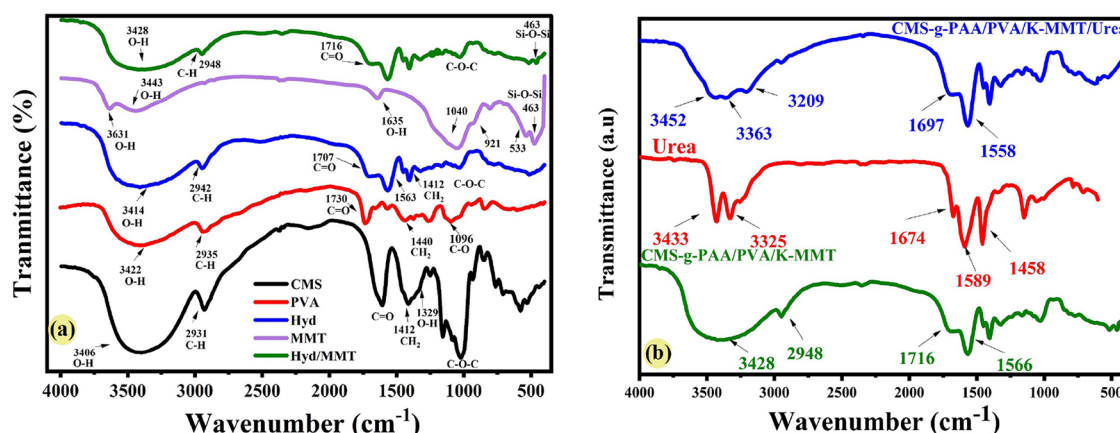


Figure 2. (a) FTIR spectra of CMS, PVA, semi IPN hydrogel and semi IPN hydrogel/K-MMT, and (b) FTIR spectra of semi IPN hydrogel/K-MMT, pure urea, and semi IPN hydrogel/K-MMT impregnated with 20 wt % of urea.

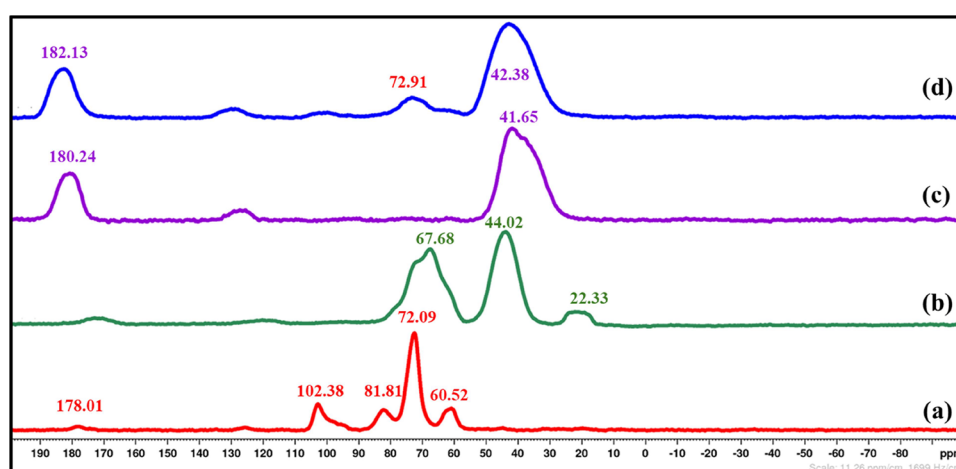


Figure 3. Solid-state ^{13}C CP-MAS NMR spectra of (a) carboxymethyl starch (b) PVA, (c) PAA, and (d) semi IPN hydrogel.

The thermal stability of native and CMS was analyzed using TGA (Figure 1d). Both exhibited two decomposition stages: water evaporation (50–120 °C) and starch backbone degradation (200–400 °C). DTG analysis (Figure 1e) showed pure starch decomposed at 314.39 °C with 78.60% weight loss, leaving an 8.41% ash residue. CMS, however, degraded at 276.05 °C with 54.32% weight loss and a higher residue (21.13%) due to carboxymethyl and sodium salt groups. The lower decomposition temperature of CMS suggests that carboxymethylation weakened the starch's crystalline structure and hydrogen bonding, reducing thermal stability, consistent with previous studies.^{42,44}

3.2. Characterization of the Semi-IPN Hydrogels.

3.2.1. Structural Analysis (FT-IR and ^{13}C NMR). Figure 2a presents the FTIR spectra of PVA, K-MMT, hydrogel, and hydrogel/K-MMT. The PVA spectrum exhibits characteristic bands at 3422 cm^{-1} (–OH stretching), 2942 and 1440 cm^{-1} (–CH₂ stretching and symmetric vibrations), 1730 cm^{-1} (C=O stretching from unhydrolyzed PVA), and 1096 cm^{-1} (C–O bonds).^{34,45} The K-MMT spectrum reveals bands at 3623 and 3430 cm^{-1} , attributed to hydroxyl group vibrations within montmorillonite and absorbed water, respectively,⁴⁶ with intensity reduction upon water absorption due to interlamellar hydrogen bonding.⁴⁷ The 1635 cm^{-1} band corresponds to water absorption, while bands at 1040 and 796 cm^{-1} represent Si–O stretching. Peaks at 918, 523, and 467 cm^{-1} are

associated with O–H deformation, Al–O–Si deformation, and Si–O–Si deformation vibrations, respectively.⁴⁸ The FTIR spectra of semi-IPN SAH show a notable decrease in the –OH band intensity, shifting to higher wavenumbers (from 3406 to 3414 cm^{-1}), indicating new inter and intramolecular hydrogen bonding. This reduction suggests the hydroxyl groups in CMS were largely consumed during grafting. A new band at 1707 cm^{-1} corresponds to the C=O stretching of AA and PVA, while the peak at 2940 cm^{-1} is attributed to AGU unit stretching in CMS and grafted PAA.⁴⁹ Further modifications are evident in the semi-IPN SAH/K-MMT composite hydrogel. The O–H stretching band shifts further to 3428 cm^{-1} , accompanied by a marked decrease in intensity, which is indicative of enhanced inter- and intramolecular hydrogen bonding between the hydroxyl groups of the polymer matrix and those on the surface of K-MMT. Additionally, the C=O stretching vibration shifts from 1707 to 1716 cm^{-1} , reflecting changes in the electronic environment of the carbonyl groups due to physical interactions with K-MMT. Notably, the appearance of a new band at 463 cm^{-1} , corresponding to the Si–O bending vibration, confirms the successful incorporation of K-MMT into the hydrogel network. These spectral changes collectively support the formation of physical cross-linking interactions between K-MMT and the hydrogel matrix, which play a significant role in enhancing the structural integrity and functional properties of the composite hydrogel. The FTIR

spectrum of CMS-g-PAA/PVA/K-MMT/urea (25 wt %) exhibits characteristic bands of both the hydrogel and urea, confirming successful impregnation (Figure 2b). Notably, the N–H stretching bands of urea (3433 and 3325 cm^{-1})⁵⁰ broadened and shifted to 3452 and 3363 cm^{-1} , while the acrylic acid carbonyl band shifted from 1716 to 1697 cm^{-1} . These spectral modifications suggest strong hydrogen-bonding interactions between urea and the polymer network, forming a highly cross-linked structure.

To further confirm the successful grafting of PAA onto the CMS backbone in the presence of PVA, ^{13}C NMR analysis was conducted on the hydrogel and hydrogel nanocomposite with K-MMT. The ^{13}C NMR spectra of these samples are shown in Figure 3. Figure 3b shows the ^{13}C CP/MAS NMR spectra of PVA, where broad peaks labeled as peaks I, II, and III are observed at approximately 77 , 71 , and 65 ppm , respectively. Peak I correspond to the formation of two intramolecular or intermolecular hydrogen bonds, peak II indicates one intramolecular or intermolecular hydrogen bond, and peak III is associated with no hydrogen bonding.⁵¹ In Figure 3c, the ^{13}C NMR spectrum of PAA shows two separate resonance peaks. The first peak, at $\delta = 180.24\text{ ppm}$, corresponds to the carbon atom in the carboxyl group ($\text{C}=\text{O}$), while the second peak, at $\delta = 41.65\text{ ppm}$, is attributed to the sp^3 hybridized carbon atoms of the $\text{CH}-\text{CH}_2$ groups in the acrylic acid backbone. The ^{13}C NMR spectrum of the semi-IPN SAH (Figure 3d) showed distinctive signals corresponding to CMS, PVA, and AA. New signals emerged at $\delta = 42.38\text{ ppm}$ and $\delta = 182.13\text{ ppm}$, which are attributed to the sp^3 hybridized carbon atoms of the $-\text{CH}_2-\text{CH}_2-$ group and the carbon atom of the $\text{C}=\text{O}$ group from the acrylic moiety, respectively. These results provide further evidence that the aliphatic chains of PAA were successfully grafted onto the CMS backbone.

3.2.2. Morphological Analysis. The surface morphology of the hydrogel samples was investigated using scanning electron microscopy (SEM), as illustrated in Figure 4. The SEM

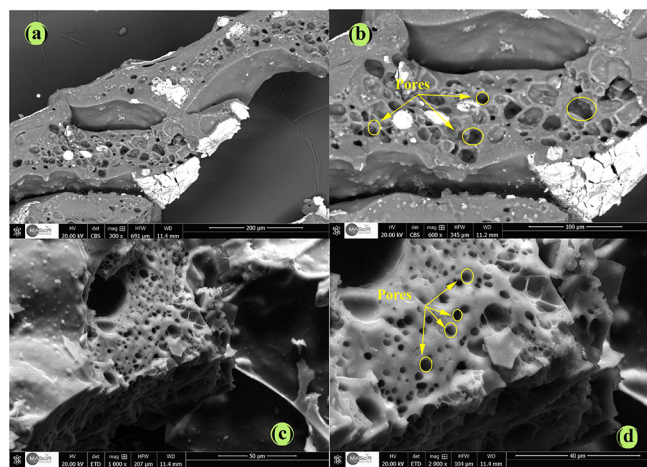


Figure 4. SEM micrographs of (a,b) semi-IPN hydrogel without K-MMT, (c,d) semi-IPN hydrogel incorporated 20 wt % K-MMT.

micrographs of the pure semi-IPN hydrogel (Figure 4a,b) reveal a honeycomb-like architecture with a homogeneous distribution of micropores, indicative of a well-organized polymer network. In contrast, the SEM images of the semi-IPN hydrogel nanocomposite incorporating K-MMT (Figure 4c,d) display a distinctly different morphology, characterized by a

highly porous structure with irregularly shaped and unevenly distributed pores. This significant alteration in morphology can be attributed to the interaction between K-MMT and the functional groups within the semi-IPN hydrogel matrix, as confirmed by FTIR and TGA analyses. The SEM images reveal that the incorporation of K-MMT enhances the structural complexity of the hydrogel network, likely functioning as a physical cross-linking agent. This phenomenon aligns with prior studies, where the incorporation of MMT resulted in increased porosity and irregular pore distribution within the hydrogel matrix.²³

3.3. Performance of the Semi-IPN Hydrogel and Semi-IPN Hydrogel Nanocomposite.

3.3.1. Effect of pH on the Swelling Property. The swelling properties of semi-IPN SAHs were assessed across a pH range of 2 to 12 (Figure 5a). Both hydrogels swelled minimally at pH 2, peaked at pH 7, and declined beyond pH 9. The nanocomposite consistently absorbed more water than the pure semi-IPN SAH. At $\text{pH} < 5$, $-\text{COO}^-$ were protonated to $-\text{COOH}$, reducing electrostatic repulsion and forming a denser, less absorbent network. Increased hydrogen bonding and cross-linking further restricted water diffusion. At pH 7, full $-\text{COOH}$ deprotonation maximized repulsion, expanding the polymer and enhancing absorption. Above pH 9, Na^+ ions shielded $-\text{COO}^-$ groups, reducing repulsion and causing shrinking of the hydrogel, thereby decreasing swelling.⁵⁰

3.3.2. Effect of Salinity on the Swelling Property. The effect of salinity on the swelling behavior of the semi-IPN hydrogel and its nanocomposite was investigated using various salt solutions (NaCl , CaCl_2 , FeCl_3) at concentrations ranging from 0.1 to 0.9 M. As the salt concentration increased, the swelling degree significantly decreased compared to distilled water (Figure 5b,c). This reduction is attributed to the changes in osmotic pressure between the polymer network and the external solution, which hindered water uptake. Additionally, the presence of multivalent cations (Ca^{2+} and Fe^{3+}) promoted complex formation with the $-\text{COO}^-$ groups in the polymer, resulting in increased cross-linking density. This enhanced cross-linking restricted the expansion of the hydrogel network and reduced its elasticity, further limiting water retention capacity. The swelling degree followed the order $\text{Na}^+ > \text{Ca}^{2+} > \text{Fe}^{3+}$, where the transition from bivalent to trivalent cations led to a higher cross-linking density due to enhanced complex formation with $-\text{COO}^-$ groups. The enhanced cross-linking reduced the elasticity of the polymer network, thereby decreasing water uptake capacity.

3.3.3. Deswelling Kinetic Studies in Saline Solutions. Swollen hydrogel samples were immersed in 0.1 M aqueous solutions of NaCl , CaCl_2 , and FeCl_3 to investigate their deswelling kinetics (Figure 5d,e). The results indicate that deswelling occurred rapidly when the swollen hydrogels were submerged in salt solutions. This behavior is attributed to the complexing capability of multivalent cations, which bonded with the carboxylate and hydroxyl groups of the polymer matrix, forming ionic cross-linking points. This interaction increased the hydrogel's cross-linking density, accelerating its deswelling. The deswelling rate of hydrogels in salt solutions increased with cationic charge. However, incorporating K-MMT into the semi-IPN hydrogel significantly slowed the deswelling rate and improved equilibrium water retention in all salt solutions compared to the semi-IPN hydrogel without K-MMT. This improvement is due to the formation of a highly

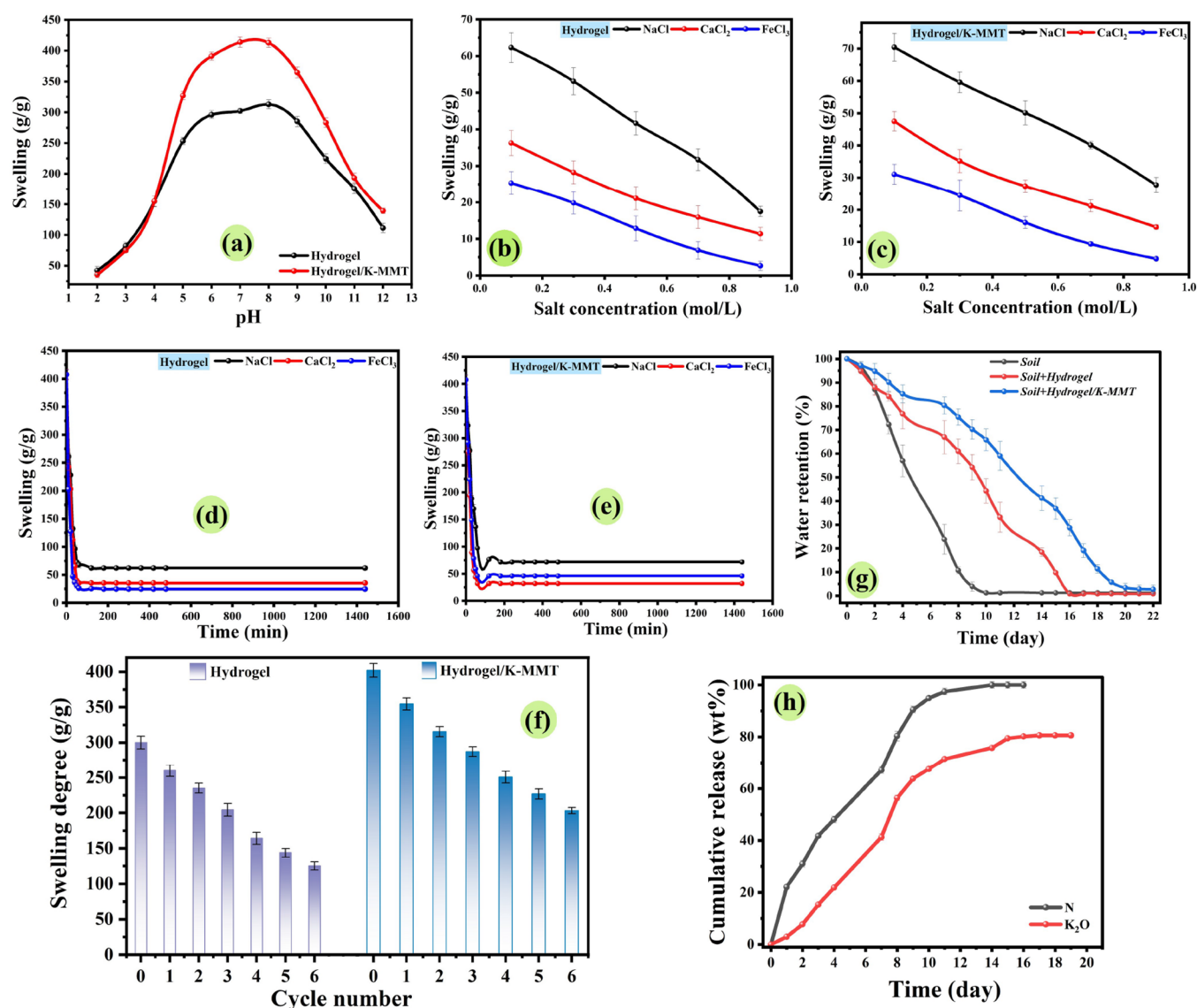


Figure 5. Effect of (a) pH on water uptake, swelling in different saline solutions of (b) semi-IPN hydrogel and (c) semi-IPN hydrogel nanocomposite. Deswelling kinetics in different saline solutions of (d) semi-IPN hydrogel and (e) semi-IPN hydrogel nanocomposite. (f) The reswelling property of the semi-IPN hydrogel and the semi-IPN hydrogel nanocomposite. (g) Water-retention behaviors of the soil with different samples. (h) cumulative nitrogen and potassium release pattern of semi-IPN hydrogel containing urea fertilizer.

cross-linked three-dimensional network with interconnected channels, which delays hydrogel shrinkage.

3.3.4. Reswelling Capability Studies. The reswelling behavior of hydrogels and their K-MMT-based nanocomposite was evaluated across successive swelling/drying cycles, as shown in Figure 5f. The swelling capacity of both hydrogels decreased progressively with each cycle. For the semi-IPN SAH composite with K-MMT, the equilibrium swelling capacity decreased from 402.03 ± 9.68 g/g in the first cycle to 203.29 ± 4.40 g/g after the sixth cycle, a reduction of 49.51%. In contrast, the semi-IPN SAP without K-MMT exhibited a greater decrease, from 300.12 ± 9.00 g/g to 125.29 ± 5.66 g/g after six cycles, corresponding to a 58.25% reduction. The superior performance of the semi-IPN hydrogel nanocomposite with K-MMT can be attributed to its enhanced structural integrity. The pure hydrogel, with its lower cross-linking density, was unable to withstand the high pressure exerted by absorbed water molecules, leading to polymer chain breakage, void collapse in the network, and a subsequent

decline in swelling capacity. In contrast, K-MMT acted as a physical cross-linker within the nanocomposite, fortifying the hydrogel network, preventing polymer chain breakage under pressure, and resulting in a more gradual reduction in swelling capacity over successive cycles.

3.3.5. Water-Retention Capacity of Soil. The water-holding capacity of sandy loam soil was assessed by watering the soil to 70% of its water-holding capacity and comparing the effects of semi-IPN hydrogel and its nanocomposite with K-MMT against a control group without hydrogel. As illustrated in Figure 5g, the control soil lost all absorbed water within 10 days, whereas the addition of hydrogel extended the water retention period to 16 days. Incorporating K-MMT into the semi-IPN hydrogel further extended this period to a maximum of 22 days, demonstrating significantly improved water retention. K-MMT acted as a cross-linker, facilitating the formation of a robust three-dimensional polymeric network with superior water-holding capacity. This performance surpasses previously reported results, showcasing higher

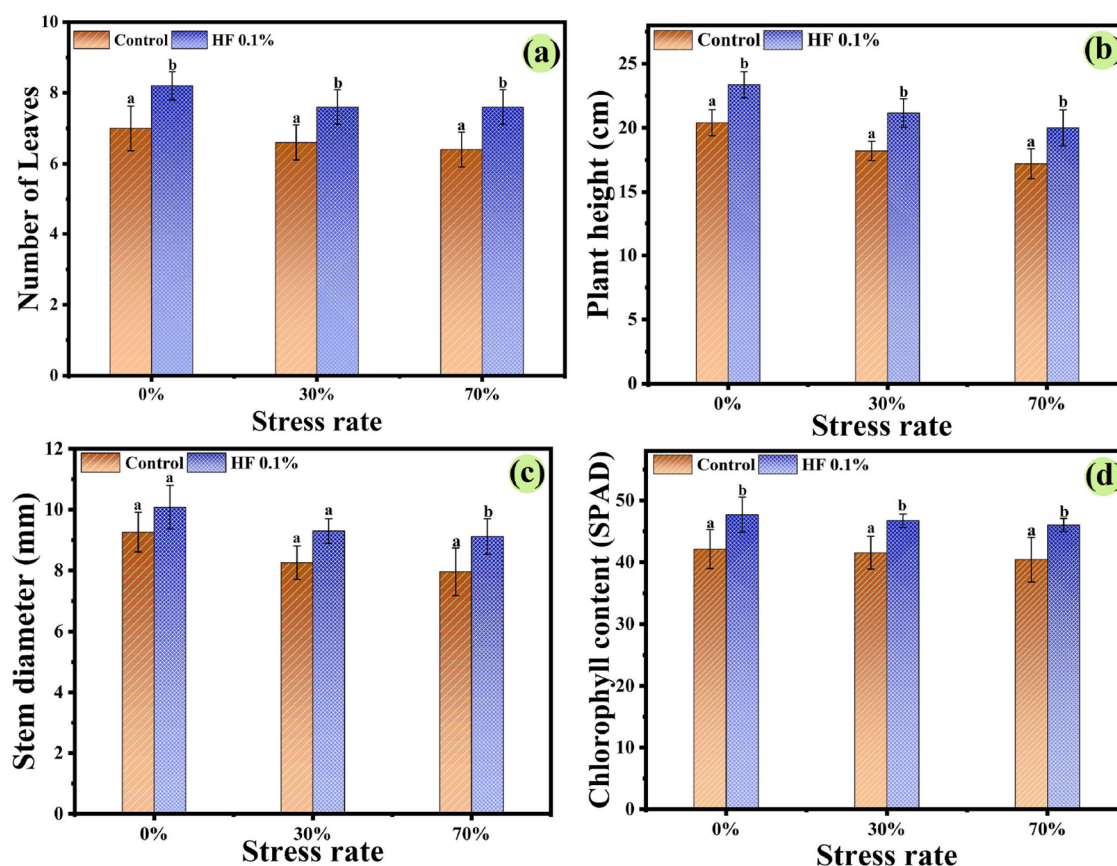


Figure 6. Interaction effects of treatments of nanocomposite (at two levels: 0 and 0.1%) and water-deficit stress (at three levels: 0, 30, and 70%) on number of leaves (a), chlorophyll content (b), length of the plant (c), and stem diameter (d).

water retention and extended retention times. Furthermore, the hydrogel not only enhances water retention but also delivers essential nutrients like nitrogen and potassium to plants. This dual functionality reduces irrigation frequency, improves soil aeration, enhances nutrient use efficiency, and bolsters crop resilience under drought conditions.

3.3.6. Slow-Release Nitrogen and Potassium in Water from SRFs. Experiments were conducted to evaluate the release profiles of N and K from the semi-IPN/K-MMT/urea fertilizer formulations by monitoring cumulative nutrient release over time (Figure 5h). The results demonstrated a slower release of N and K compared to commercial urea and potassium fertilizers, which fully dissolve in distilled water within a day.³³ During the first 24 h, the formulations exhibited initial release rates of 22.15% for N and 2.95% for K. Notably, the total release of K remained below 80%, which was lower than that of N. The CMS-g-PAA/PVA/K-MMT system, a slightly cross-linked hydrophilic polyelectrolyte, facilitated water penetration via capillary action, causing the dissociation of hydrophilic groups in the polymer chain upon immersion. This dissociation generated negative charges, leading to electrostatic repulsion and swelling of the polymer network. Fertilizer release occurred as trapped nutrients first dissolved within the swollen gel and then diffused out due to concentration gradients. This diffusion process was slowed by adsorption and steric effects of the swollen gel, which influenced nitrogen and potassium release differently. The nitrogen was present in two forms: as amides bonded to the CMS and AA copolymer chain, and as biuret compounds incorporated into the composite. Nitrogen release occurred

gradually, as biuret compounds and amides dissolved sequentially, with steric hindrance and adsorption effects in the swollen gel further slowing its diffusion. Conversely, potassium, incorporated through neutralization of PAA and interaction with MMT, had no chemical bonds with the polymer backbone. The positive potassium ions were retained by electrostatic attraction within the semi-IPN hydrogel network, resulting in slower and lower total K release compared to nitrogen. Although the urea release rate in our formulation (more than 90% over 10 days) is relatively faster than that of traditional hydrophobic-coated systems, this release profile is moisture-responsive and tailored to match the early nutrient demands of crops, which is especially beneficial for short-cycle cultivation and drought-prone environments. In addition to nutrient release, our hydrogel also functions as a soil moisture reservoir, addressing two critical agronomic challenges simultaneously. By contrast, hydrophobic-coated fertilizers are typically designed only for nutrient delivery and do not contribute to water retention, limiting their overall effectiveness in dry conditions. Furthermore, these systems are generally intended for single-use applications, which increases input costs over multiple cropping cycles. Taken together, our hydrogel system offers a more practical and multifunctional solution, combining efficient nutrient release with prolonged water retention to enhance plant performance under water stress.

3.4. Comparison with Previously Reported Super-absorbent Systems in the Literature. The performance of our hydrogel was evaluated in comparison to various hydrogels reported in the literature, focusing on water absorption

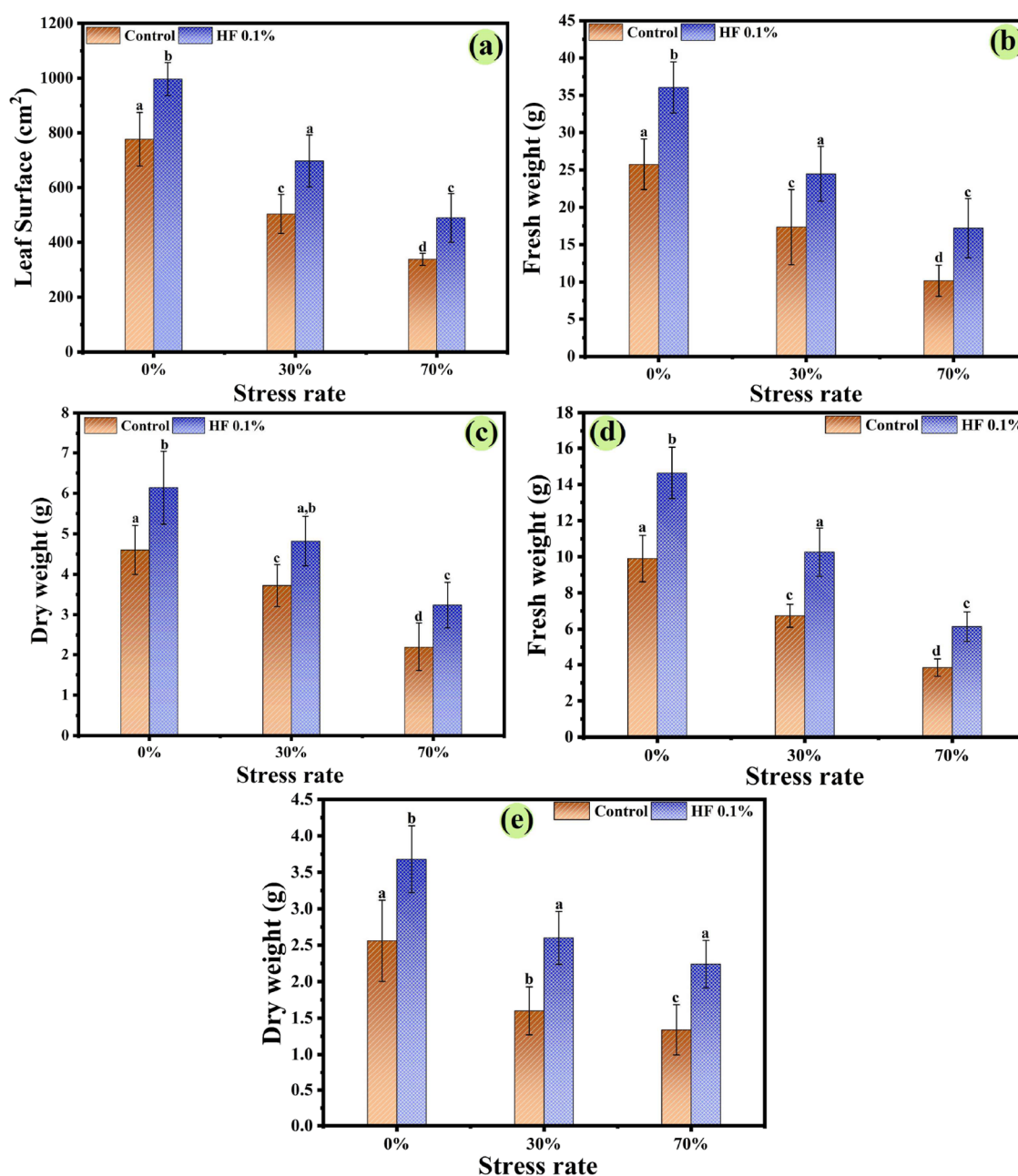


Figure 7. Interaction effects of treatments of nanocomposite (at two levels: 0 and 0.1%) and water-deficit stress (at three levels: 0, 30, and 70%) on leaf area (a). Interaction effects of treatments of nanocomposite (at two levels: 0 and 0.1%) and water-deficit stress (at three levels: 0, 30, and 70%) on Above ground fresh and dry matter (b,c), fresh and dry root matter (d,e).

capacity and soil water retention (Table S3). As shown, our hydrogel demonstrates strong competitiveness in both aspects. For instance, Chitosan-g-poly(acrylic acid) exhibits an absorption capacity of approximately 159.92 g/g and a soil water retention capacity of 11.7% after 7 days with 1% hydrogel application.⁵² Similarly, the Starch-g-poly(AA-co-AM)/ modified natural char nanoparticles hydrogel shows a water absorption of 389.9 g/g and a soil water retention of 7.15% after 14 days at 1 wt % hydrogel content.⁵³ Our hydrogel demonstrates a superior water absorption capacity of 392.87 ± 8.05 g/g and extends soil water retention up to 22 days. This combination of high swelling capacity and prolonged water release under soil conditions underscores the efficacy and practical potential of our hydrogel system for sustainable water management in agricultural applications.

3.5. Effect of Semi-IPN Hydrogel on Maize Growth Parameters.

3.5.1. Effect of Semi-IPN Hydrogel on Leaf Number. The results showed that stress levels did not significantly affect leaf numbers ($p = 0.161$ and 0.105 for 0 and 0.1% SAH doses, respectively (Figure 7a). However, treatments with the semi-IPN hydrogel nanocomposite exhibited significantly higher leaf numbers than the control ($p = 0.009$, 0.02 , and 0.012 for 70, 30, and 0% stress levels, respectively). This enhancement is attributed to the hydrogel's high water-holding capacity, enabling gradual water release and maintaining soil moisture, even under 70% water stress. CMS-based hydrogels ensure steady water and nutrient availability, benefiting plant growth in arid regions. Similar findings by Al-Jabari et al.⁵⁴ showed that SAH from used diapers improved tomato growth, with optimal results at a 2% hydrogel addition,

while in our work, we achieved significant growth improvements using only 0.1%.

3.5.2. Effect of Semi-IPN Hydrogel on Plant Height. Figure 6b illustrates the effects of water stress and hydrogel application on maize plant height in sandy-loam soil. While water stress reduced plant height by 10.78 and 15.68% at stress levels of 30 and 70%, respectively, compared to unstressed plants, these reductions were not statistically significant ($P > 0.05$). The addition of 0.1% semi-IPN/K-MMT/urea hydrogel significantly mitigated these effects, increasing plant height by 12.67, 13.98, and 14.32% under stress levels of 0, 30, and 70%, respectively, with P -values of 0.003, 0.002, and 0.016. The hydrogel's ability to retain and gradually release water maintained consistent soil moisture, enabling better plant growth even under water-stressed conditions. This demonstrates the potential of hydrogels to improve agricultural productivity, particularly in water-limited environments.

3.5.3. Effect of Semi-IPN Hydrogel on Stem Diameter. Figure 7c shows the effect of water stress and hydrogel application on the stem diameter of maize plants in sandy-loam soil. No significant difference was observed between the hydrogel and the control under 0 and 30% stress, but a significant improvement was noted under 70% stress ($P = 0.037$). Plants treated with 0.1% hydrogel showed an 11.18% increase in stem diameter under 30% stress and 12.72% under 70% stress compared to the control. This improvement is attributed to the hydrogel's swelling and water retention capacity, which gradually releases nutrients such as urea and potassium, supporting plant growth and resilience in water-limited conditions. These findings align with research by Madramootoo et al.⁵⁵ and Hernández-Hernández et al.,⁵⁶ who observed similar effects with hydrogels from wastepaper and chitosan, respectively.

3.5.4. Effect of Semi-IPN Hydrogel on Chlorophyll Content. The total chlorophyll levels showed no significant differences between drought stress treatments (Figure 6d). However, treatments with the semi-IPN hydrogel nanocomposite significantly improved chlorophyll content compared to the control group, with p -values of 0.018, 0.014, and 0.03 for 70, 30, and 0% stress levels, respectively. Chlorophyll content increased by 2% under normal irrigation, 16.74% under 30% stress, and 30% under 70% stress with hydrogel treatment. The application of SAH to the soil yielded positive results, resulting in a significant increase in chlorophyll content compared to the control treatment. This increase in chlorophyll content after application of the SAH may be attributed to a reduction in chlorophyll degradation or an increase in chlorophyll biosynthesis due to the supply of water and nutrients to tomato plants grown under water-deficient conditions. The observed correlation between chlorophyll content, drought stress and hydrogel application aligns with the results of various studies suggesting that hydrogel positively influences photosynthetic activity by increasing water availability to the plant.^{57–59}

Soil moisture was assessed following the vegetative stage of Maize and prior to harvest (Table S3). Hydrogel-treated soils exhibited a notable retention of moisture, surpassing that of untreated controls, particularly under conditions of moderate to severe water stress. This finding substantiates the efficacy of hydrogels in curtailing moisture loss and enhancing soil water retention. By mitigating the repercussions of diminished irrigation, hydrogels augment water use efficiency, rendering

them a promising soil amendment for sustainable agriculture in regions afflicted by drought.

3.5.5. Effect of Semi-IPN Hydrogel on Leaf Area (LA). Leaf area was measured using the AM350 hand-held device (Figure 7a). Drought stress significantly reduced leaf area ($p < 0.001$), due to due to impaired mineral nutrition, metabolism, and stomatal closure, which limited CO_2 availability and photosynthesis.⁶⁰ This finding is consistent with the results of previous studies that reported similar reductions in leaf area under drought stress.^{61–63} However, applying 0.1% semi-IPN hydrogel significantly increased leaf area under water stress ($P = 0.006$), with a 27.83% rise at 30% stress and 30.92% at 70% stress. The observed benefits of hydrogel application result from its ability to increase soil water retention, improve accessibility to water and nutrients, and reduce the production of reactive oxygen species (ROS), collectively enhancing plant resistance to drought stress.⁶⁴ The ability of hydrogels to sustain photosynthesis and maintain turgor pressure suggests their potential as an effective strategy for mitigating drought stress and enhancing crop resilience in arid environments.

3.5.6. Effect of Semi-IPN Hydrogel on Above-Ground Fresh and Dry Matter. The application of hydrogels resulted in a significant increase in the surface fresh weight of the maize plant under both normal and drought conditions ($P < 0.001$), with an increase of 14.30% ($P = 0.003$) under optimal irrigation, 29.16% ($P = 0.044$) at 30% stress, and 40.93% ($P = 0.014$) at 70% stress (Figure 7b). Similarly, above-ground dry weight increased by 25.08% ($P = 0.022$) under optimal irrigation, 22.82% ($P = 0.025$) at 30% stress, and 32.09% ($P = 0.035$) at 70% stress (Figure 7c). These findings are consistent with those reported by Abdallah et al.,⁶⁵ The increased biomass is attributed to improved soil moisture retention and nutrient availability, supporting plant physiological processes under drought. These findings underscore hydrogel's effectiveness in mitigating drought stress and enhancing maize productivity in both optimal and water-stressed conditions.

3.5.7. Fresh and Dry Root Matter. The effect of irrigation levels and semi-IPN hydrogel application on root biomass was statistically significant ($P < 0.001$ for fresh weight; $P = 0.004$ for dry weight). Plants receiving 100% irrigation exhibited the highest root biomass, followed by those at 70% and 30% irrigation levels. Hydrogel-treated plants showed significantly larger roots ($P < 0.001$), with an increase of 34.30% under 30% water stress ($P = 0.001$) and 37.45% under 70% water stress ($P = 0.001$). Dry root weight increased by 38.46% at 30% stress and 40.17% at 70% stress ($P = 0.004$, $P = 0.005$) (Figure 7e). By improving soil moisture availability, semi-IPN hydrogel supports root growth, enhances water and nutrient uptake, and strengthens plant resilience to water stress.

Drought stress limits plant water and nutrient uptake, causing delayed growth, reduced root length, and lower biomass, potentially leading to plant death. A strong root system is crucial for water and nutrient absorption and can help mitigate drought stress. This study shows that applying a CMS-based hydrogel significantly improved soil water availability at the root level of maize plants under water stress. Despite typical biomass reduction under limited irrigation, hydrogel-treated plants showed increased fresh root weight due to enhanced soil water retention. Additionally, the hydrogel improved soil porosity and aeration, promoting root development. These findings align with previous research by Goma and Aldaby,⁶⁶ supporting the effectiveness of hydrogels in enhancing plant growth under drought conditions.

3.5.8. N, P, and K Uptake. Water stress significantly reduces the uptake of essential nutrients (N, P, K), negatively impacting plant growth. However, as shown in Table S3, the application of 0.1%wt of semi-IPN hydrogels significantly improved nutrient absorption ($P < 0.05$). Under 30–70% water stress, hydrogel incorporation increased N uptake by 103.87–262.83%, P by 91.3–476.51%, and K by 105.25–207.16%, leading to higher fresh and dry biomass. These findings align with Liang and Shi,⁶⁷ who reported enhanced nutrient uptake efficiency with poly- γ -glutamic acid. The observed improvements result from (i) controlled nutrient release, ensuring steady availability; (ii) improved soil moisture retention, facilitating nutrient absorption; and (iii) enhanced soil structure, promoting root growth.^{68–71} These mechanisms highlight the hydrogel's potential in mitigating drought-induced nutrient deficiencies, supporting plant health, and ensuring stable yields in water-limited environments.

Although the swelling behavior of the developed hydrogel was primarily evaluated under controlled laboratory conditions, the results demonstrated excellent water absorption and retention capacity. However, for practical agricultural applications, especially in regions with diverse and fluctuating climatic conditions, it is essential to assess hydrogel performance across a broad range of humidity levels. For instance, Zhou et al. developed super moisture-absorbent gels (SMAGs) that could harvest approximately 4 g/g of water at 90% relative humidity and sustain plant survival for over 14 days without liquid irrigation. Notably, their SMAG-based system maintained a stable internal humidity ($\sim 70\%$) despite wide external fluctuations ranging from 30 to 90% RH, effectively mimicking transitions from arid to humid environments.⁷² Similarly, Bai et al. reported a biodegradable lignin–chitosan– CaCl_2 hydrogel capable of absorbing atmospheric moisture across a wide humidity range (30–90% RH), with a maximum absorption of 2.03 g/g at 90% RH and 0.43 g/g even at 30% RH.⁷³ This hydrogel demonstrated efficient water release and long-term durability, significantly improving soil water retention under both low and high humidity conditions. These studies illustrate the adaptability of hydrogels to different environmental humidity levels and their potential role in sustainable water management. In our study, the agronomic evaluation of the hydrogel under varying water stress conditions, normal irrigation, mild stress, and severe stress, was designed to simulate such humidity variability. The observed enhancement in maize growth under all tested conditions further reinforces the hydrogel's multifunctional performance and suitability for diverse agricultural environments.

4. CONCLUSIONS

In this study new semi-IPN hydrogel nanocomposites were successfully developed by graft copolymerizing acrylic acid monomers onto a CMS backbone in the presence of PVA chains, with K-MMT as a filler, KPS as an initiator, and MBA as a cross-linking agent in aqueous solution. We investigated different concentrations of components to identify the optimal composition for maximum water absorption. The optimized composite, containing 20 wt % K-MMT, exhibited a remarkable water absorption capacity of 392.87 ± 8.05 g/g in deionized water. The swelling behavior of the composite was significantly influenced by factors such as salt type, concentration, and pH of the external solution. In soil, the hydrogel demonstrated excellent water retention properties, the optimized semi-IPN hydrogel nanocomposite maintaining

soil moisture for over 23 days, effectively reducing the frequency of irrigation. Furthermore, the hydrogel exhibited slow-release properties for nitrogen and potassium fertilizers, providing a steady nutrient supply to plants. The impact of water stress on corn growth was significant, with reduced growth observed under drought conditions. However, the application of the semi-IPN hydrogel significantly improved plant growth parameters, including leaf count, chlorophyll content, stem diameter, leaf area, and both fresh and dry matter production. The 0.1% semi-IPN hydrogel treatment also enhanced fertilizer uptake efficiency, demonstrating the hydrogel's ability to improve soil physical and chemical properties, such as water retention capacity, adsorption, and nutrient availability. These results indicate that the semi-IPN nanocomposite hydrogel is a promising material for improving plant growth and nutrient uptake under drought stress. However, further field trials on different crops are needed to validate their effectiveness in real agricultural settings. Additionally, future studies should evaluate the economic feasibility of using this hydrogel on a large scale, considering the potential for reduced water consumption and increased crop yields.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acssuschemeng.5c02331>.

Details of the preparation of semi-IPN SAHs, swelling measurements, TGA characterization, swelling kinetics, and pot culture experiments on maize plants (PDF)

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